WINDOWS SERVER CONFIGURATION REPORT

Active Directory • DNS • DHCP • Group Policy • IIS

Domain: trushti.local
Server: TRUSHTI-SRV01

1. Introduction

This report presents a Windows Server environment configured around the domain trushti.local. The lab covers Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy and Internet Information Services (IIS). These services are commonly combined to provide centralized identity management, network configuration, policy administration and web hosting.

2. Lab Configuration

Parameter	Configuration
Operating System	Windows Server 2022
Server Name	TRUSHTI-SRV01
Domain	trushti.local
Server IP	192.168.10.10
DHCP Scope	192.168.10.100 – 192.168.10.200
Gateway	192.168.10.1
DNS Server	192.168.10.10
IIS Binding	http://trushti.local

3. Active Directory Domain Services (AD DS)

AD DS stores information about users, computers, groups and other directory objects. It provides centralized authentication and authorization in a Windows domain.

Implementation: Install the AD DS role, promote TRUSHTI-SRV01 to a domain controller, create the new forest trushti.local, complete the prerequisite checks and restart the server. Users and groups can then be managed through Active Directory Users and Computers.

4. DNS

DNS resolves host and domain names to IP addresses. Active Directory relies heavily on DNS for locating domain controllers and other services.

Implementation: Configure an Active Directory-integrated forward lookup zone named trushti.local and verify that the server's A record resolves to 192.168.10.10.

Verification commands: nslookup trushti.local | nslookup TRUSHTI-SRV01.trushti.local

5. DHCP

DHCP automatically assigns IP configuration to clients. A scope defines the pool of addresses available for leases and can also provide gateway and DNS settings.

Example scope: 192.168.10.100–192.168.10.200, subnet mask 255.255.255.0, gateway 192.168.10.1, DNS 192.168.10.10, DNS domain trushti.local.

Verification command on a client: ipconfig /all

6. Group Policy

Group Policy provides centralized control of user and computer configuration. A Group Policy Object (GPO) can be linked to the domain or an organizational unit.

Example: create a GPO named Trushti Security Policy, link it to the required domain/OU, configure appropriate security or desktop settings, then verify with gpupdate /force and gpresult /r.

7. IIS

Internet Information Services (IIS) is Microsoft's web-server platform for Windows. It can host websites and web applications using HTTP or HTTPS.

Implementation: install the Web Server (IIS) role, create a website named Trushti Website, use a physical path under C:\inetpub\wwwroot\trushti, and configure an HTTP binding on port 80. The site can be tested with http://trushti.local.

8. Illustrative Windows Server Screenshots

The following composite visual represents the administrative-console screens expected after completing the configuration.

Windows Server Configuration Report

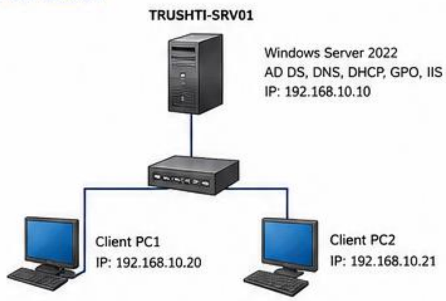
Configured Active Directory, DNS, DHCP, Group Policy, IIS in a Windows Server Environment

Domain Name	trushti.local
Environment	Windows Server 2022 Standard Evaluation
Server Name	TRUSHTI-SRV01
Domain Controller IP	192.168.10.10
Date	30 May 2025

1. INTRODUCTION

In this lab, a Windows Server 2022 environment was configured by implementing core server roles and services including Active Directory Domain Services (AD DS), DNS, DHCP, Group Policy, and IIS.
The domain name used is *trushti.local*.

2. LAB TOPOLOGY



3. ACTIVE DIRECTORY DOMAIN SERVICES (AD DS)

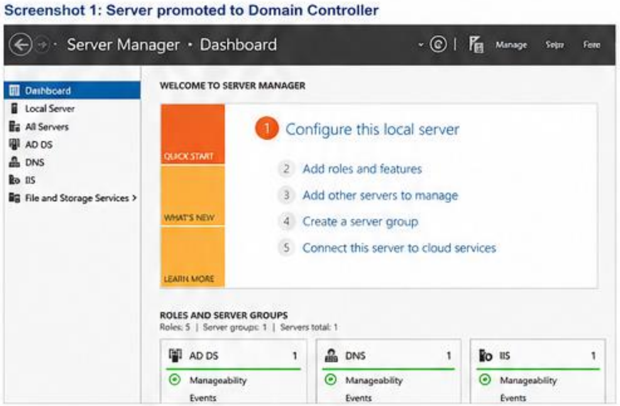
Theory:

Active Directory Domain Services stores information about network objects and makes this information available to network users and administrators. It authenticates users and computers on the network.

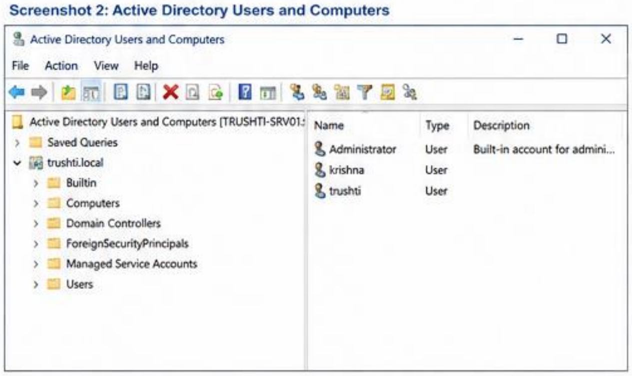
Implementation Steps:

1. Installed AD DS role.
2. Promoted the server to a Domain Controller.
3. Created a new forest: *trushti.local*.
4. Rebooted the server.

Screenshot 1: Server promoted to Domain Controller



Screenshot 2: Active Directory Users and Computers



4. DNS (DOMAIN NAME SYSTEM)

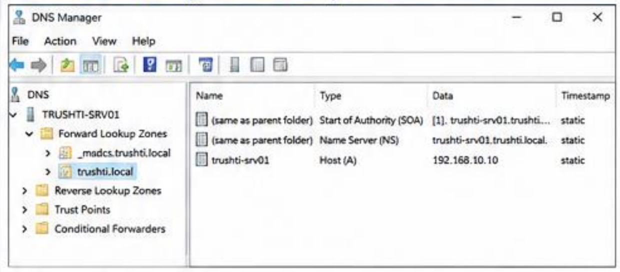
Theory:

DNS translates domain names into IP addresses. Since AD DS is installed, DNS role was installed to resolve names within the domain.

Implementation Steps:

1. Installed DNS Server role.
2. DNS automatically integrated with AD.
3. Verified forward lookup zone: *trushti.local*.

Screenshot 3: DNS Manager – Forward Lookup Zone



5. DHCP (DYNAMIC HOST CONFIGURATION PROTOCOL)

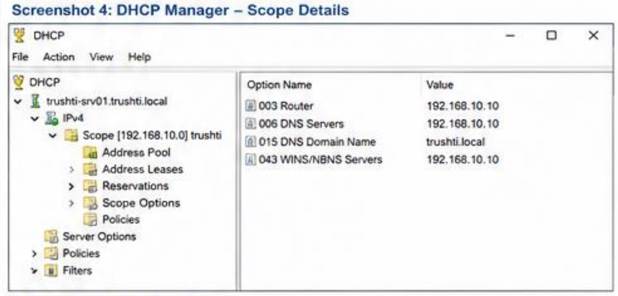
Theory:

DHCP automatically assigns IP addresses to devices on the network.

Implementation Steps:

1. Installed DHCP Server role.
2. Created a new scope: 192.168.10.0/24.
3. Excluded IP range: 192.168.10.1 – 192.168.10.10.
4. Scope range: 192.168.10.20 – 192.168.10.200.
5. Set router (default gateway): 192.168.10.10.
6. Set DNS Server: 192.168.10.10.

Screenshot 4: DHCP Manager – Scope Details



6. GROUP POLICY (GPO)

Theory:

Group Policy allows administrators to manage user and computer settings in the domain.

Implementation Steps:

1. Created a GPO to set desktop wallpaper for domain users.
2. Linked the GPO to the domain.

Screenshot 5: Group Policy Management – GPO Created

7. IIS (INTERNET INFORMATION SERVICES)

Theory:

IIS is a web server used to host websites and web applications.

Implementation Steps:

1. Installed IIS Web Server role.
2. Created a default website.
3. Verified the website using a browser.

Screenshot 6: IIS Manager – Default Website Running

8. RESULT

All required services (AD DS, DNS, DHCP, Group Policy, IIS) were successfully installed and configured in the Windows Server environment.

9. CONCLUSION

This lab demonstrates the integration of essential Windows Server services to provide a complete network infrastructure for an organization using the domain *trushti.local*.

Figure 1 — Illustrative Server Manager, AD Users and Computers, DNS Manager, DHCP Manager, Group Policy Management and IIS Manager views.

9. Verification Checklist

Service	Test	Expected Result
AD DS	Open Active Directory Users and Computers	trushti.local objects are visible
DNS	nslookup trushti.local	Name resolves correctly
DHCP	ipconfig /all on client	Client receives an address from the scope
Group Policy	gpupdate /force; gpresult /r	Configured GPO is applied
IIS	Open http://trushti.local	Website loads successfully

10. Result

The planned Windows Server environment contains the five requested services and demonstrates how they work together. AD DS provides centralized identity, DNS supports domain name resolution, DHCP automates network configuration, Group Policy provides centralized administration, and IIS provides web hosting.

11. Conclusion

This practical provides an enterprise-style introduction to Windows Server administration. The trushti.local domain acts as the central identity namespace, while DNS, DHCP, Group Policy and IIS provide complementary infrastructure services.